

Individual_Assignment_5

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1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# reading .xlsx files
library(readxl)

# wrap axis labels
library(scales)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  # powerful argument in read_xlsx to show the missing variables as numbers
  na = c("", "n/a", "N/A", "over", "173+"))

```

Class Code

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

# creating new object from taxon list
taxon_list_clean <- taxon_list |>
  # converted tacon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

```

```

water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>

```

```

# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new object
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(
  # columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)

```

```

summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# created new column to join with data frames
unite("date_site", date, site, remove = FALSE) |>
# join with water_quality_clean
left_join(water_quality_clean, by = "date_site") |>
# joined with taxon_list_clean
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "East Channel",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full,
                               med_sal,
                               .fun = "median",
                               .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

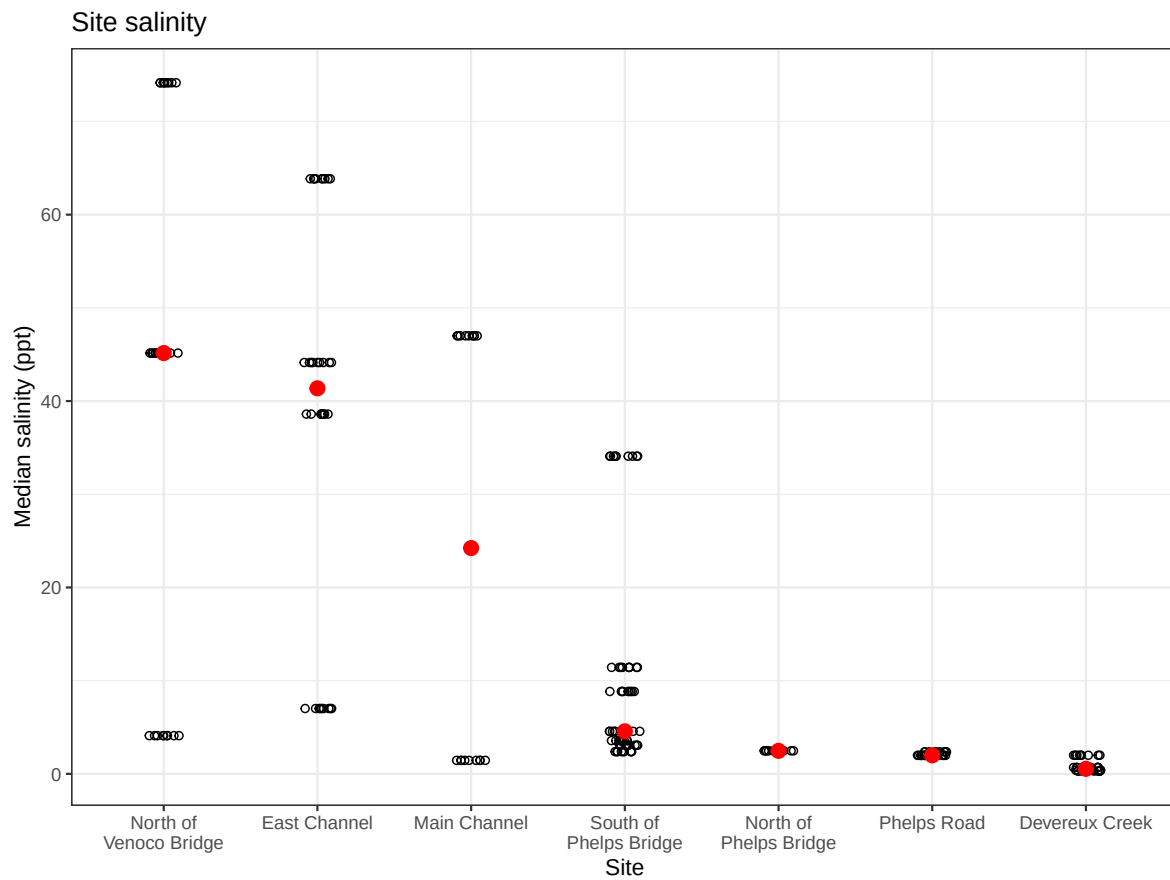
```

# base layer
salinity_plot <- ggplot(data = aq_ins_clean,
  # x-axis: full site name
  # y-axis: median salinity
  mapping = aes(x = site_full,
                y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# points representing median salinity
stat_summary(geom = "point",
            fun = median,
            color = "red",
            size = 3) +

```

```
# wrapping x-axis labels
scale_x_discrete(labels = label_wrap(width = 15)) +
# relabeling axes, adding title
labs(x = "Site",
      y = "Median salinity (ppt)",
      title = "Site salinity") +
# using different theme from default
theme_bw()

# displaying plot
salinity_plot
```



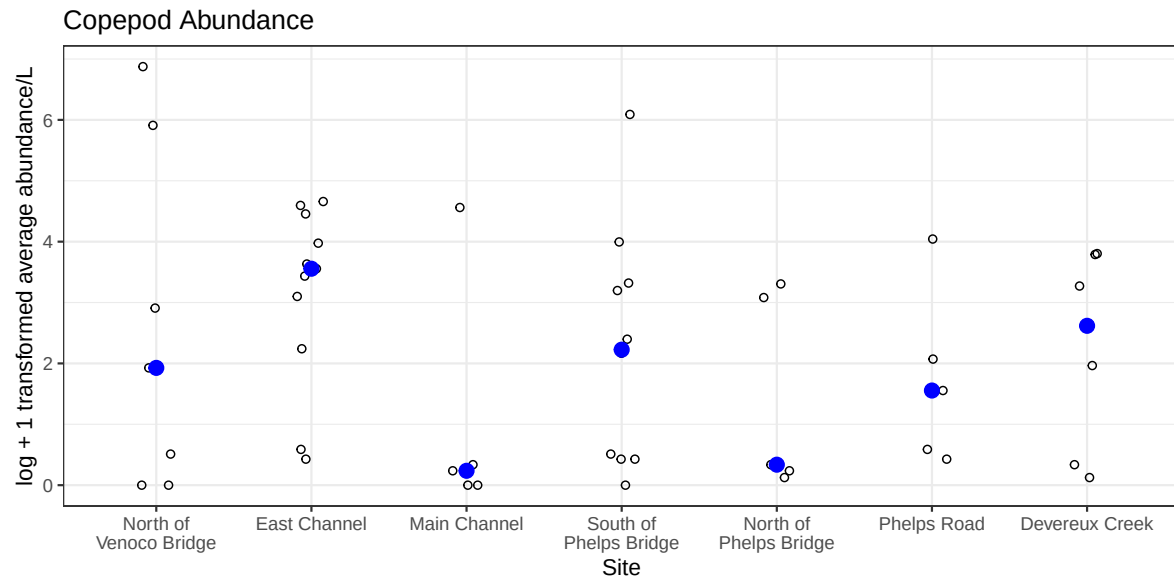
```
copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

```

# base layer, creating oobject to store plot
copepod_plot <- ggplot(data = copepods,
  # x-axis: full site name
  # y-axis: log transformed average count/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
# first layer: points to represent observations, jitter to represent each
↪ sample
geom_jitter(height = 0,
  shape = 21,
  width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
  fun = "median",
  size = 3,
  color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
  y = "log + 1 transformed average abundance/L",
  title = "Copepod Abundance") +
# using different theme
theme_bw()

# displaying object
copepod_plot

```



`salinity_plot` / `copepod_plot`

Scatter plot showing the log + 1 transformed average abundance/L for various sites. The y-axis ranges from 0 to 6. The x-axis lists seven sites: North of Venoco Bridge, East Channel, Main Channel, South of Phelps Bridge, North of Phelps Bridge, Phelps Road, and Devereux Creek. Data points are represented by open circles, with some points highlighted in blue.

Site	log + 1 transformed average abundance/L	Highlighted
North of Venoco Bridge	0.0	No
North of Venoco Bridge	0.5	No
North of Venoco Bridge	2.9	No
North of Venoco Bridge	5.9	No
North of Venoco Bridge	6.8	No
North of Venoco Bridge	1.9	Yes
East Channel	0.5	No
East Channel	0.4	No
East Channel	2.2	No
East Channel	3.5	No
East Channel	3.6	Yes
East Channel	4.0	No
East Channel	4.6	No
East Channel	4.7	No
East Channel	3.1	No
Main Channel	0.0	No
Main Channel	0.2	No
Main Channel	0.3	No
Main Channel	0.2	Yes
Main Channel	4.6	No
South of Phelps Bridge	0.0	No
South of Phelps Bridge	0.5	No
South of Phelps Bridge	0.4	No
South of Phelps Bridge	2.2	No
South of Phelps Bridge	2.3	Yes
South of Phelps Bridge	2.4	No
South of Phelps Bridge	3.3	No
South of Phelps Bridge	3.2	No
South of Phelps Bridge	4.0	No
South of Phelps Bridge	6.1	No
North of Phelps Bridge	0.2	No
North of Phelps Bridge	0.3	No
North of Phelps Bridge	3.1	No
North of Phelps Bridge	3.3	No
North of Phelps Bridge	0.3	Yes
Phelps Road	0.4	No
Phelps Road	1.5	No
Phelps Road	1.5	Yes
Phelps Road	2.1	No
Phelps Road	4.0	No
Devereux Creek	0.3	No
Devereux Creek	0.1	No
Devereux Creek	2.0	No
Devereux Creek	3.3	No
Devereux Creek	2.6	Yes
Devereux Creek	3.8	No
Devereux Creek	3.8	No

Problems

1. Visualizing differences across sites in major taxon abundance

In class, we visualized the differences across sites in median (log transformed) copepod abundance.

In this section, you will create another figure to add to the multipanel plot of salinity and copepod abundance to show ostracod abundance.

a. Plan your figure

- x-axis: `site_full`
- y-axis: `log_ave_lit`
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
ostracod <- aq_ins_clean |>
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

c. Code your figure

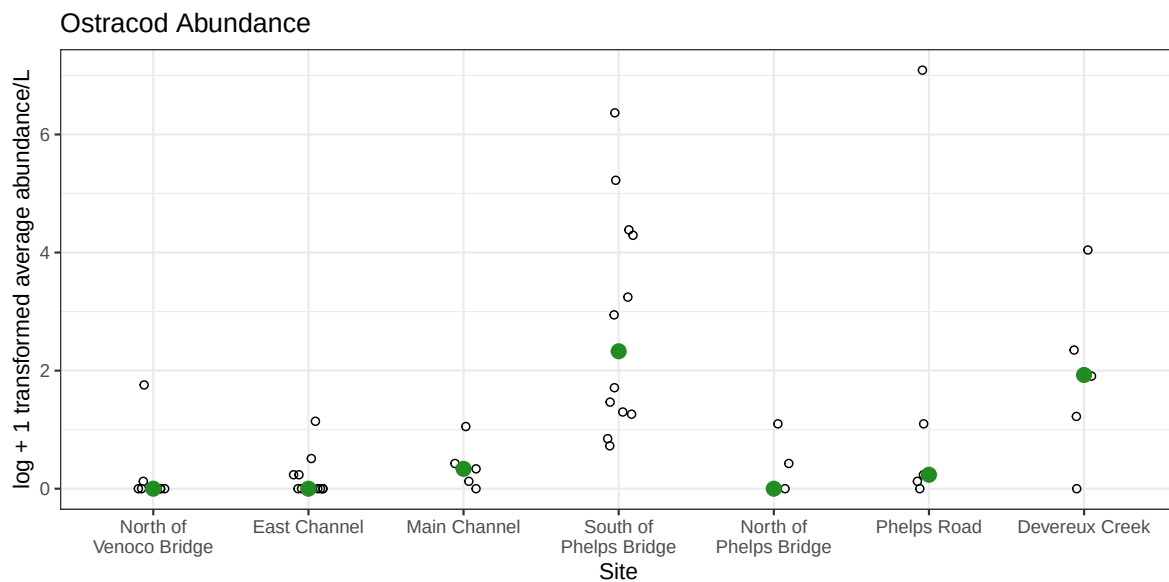
```
# base layer, creating oobject to store plot
ostracod_plot <- ggplot(data = ostracod,
  # x-axis: full site name
  # y-axis: log transformed average count/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations, jitter to represent each
  # sample
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # second layer: points to represent medians
```

```

stat_summary(geom = "point",
             fun = "median",
             size = 3,
             color = "forestgreen") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracod Abundance") +
# using different theme
theme_bw()

# displaying object
ostracod_plot

```



d. Create a multipanel figure

```

salinity_plot / ( copepod_plot | ostracod_plot )

```

